Lecture 6 Scribe

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I want you to generate a lecture scribe for Lecture 6 that I can use as exam reference material. The scribe should be an exact reformation of what was taught in the lecture, so that I can revise the lecture only from this document. Please use only the provided lecture slides and any textbook content that is directly part of those slides. Do not add explanations, examples, intuition, or material that is not present in the lecture. lect01 is one example lecture scribe. This example is provided just to show the expected structure and organization of a lecture scribe. Its content is not at all related to Lecture 6 and should not affect the actual material written. Example Lecture Scribe (structure only): Lecture __ 1. Overview Brief outline of the topics covered in the lecture. 2. Definitions and Notation All definitions, symbols, and assumptions introduced in the lecture. 3. Main Results Statements of theorems, propositions, or results as presented. 4. Proofs or Derivations Step-by-step proofs or derivations exactly as shown in the lecture, including stated assumptions and intermediate steps. 5. Examples Examples discussed in the lecture, written step by step without adding or removing steps. 6. Summary Short recap of the lecture content. Using the same structure as the example above, generate a lecture scribe for Lecture 6. The scribe should be organized clearly for exam preparation and must include definitions, assumptions, proofs, derivations, and worked examples only if they are present in the provided lecture material. Do not invent or infer any content beyond what is given.

Lecture 6 Scribe – Discrete Random Variables And Expectation

Lecture 6

Discrete Random Variables, Expectation, and Problem Solving

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1. Overview

This lecture covers the concept of random variables with emphasis on discrete random variables. Topics include probability mass functions (PMFs), cumulative distribution functions (CDFs), Bayes' Theorem, independent events, and standard types of discrete random variables such as Bernoulli, Binomial, Geometric, and Poisson random variables, along with worked examples presented in the lecture.

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2. Definitions and Notation

Random Variable

A random variable X on a sample space Ω is a function:

$X : \Omega \to \mathbb{R}$

assigning a real number to each sample point $\omega \in \Omega$.

In this lecture, attention is restricted to **discrete random variables**, i.e., random variables that take values in a finite or countably infinite subset of $\mathbb{R}$.

Probability Mass Function (PMF)

A random variable that can take on at most a countable number of possible values is said to be discrete.

Let X be a discrete random variable with range $Rx = \{x_1, x_2, x_3, \ldots\}$. The function:

$P_x(x_k) = P(X = x_k)$, $k = 1,2,3,\ldots$

is called the **Probability Mass Function (PMF)** of X.

Since X must take on one of its possible values:

$\Sigma_k P_x(x_k) = 1$

Cumulative Distribution Function (CDF)

The cumulative distribution function of a random variable X is defined as:

$F_x(x) = P(X \leqslant x)$

(Properties and examples were discussed in the lecture.)

Bayes' Theorem

Using:

$P(A \cap B_i) = P(B_i \mid A)P(A)$

we get:

$P(B_i \mid A) = [P(A \mid B_i)P(B_i)] / [\Sigma_j P(A \mid B_j)P(B_j)]$

This is known as **Bayes' Formula**.

- $P(B_i)$: a priori probability
- $P(B_i \mid A)$: posteriori probability

Independent Events

Two events A and B are independent if:

$P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$

Equivalently:

$P(A \cap B) = P(A)P(B)$

For three events A, B, and C, they are **mutually independent** if:

$P(A \cap B) = P(A)P(B)$ $P(A \cap C) = P(A)P(C)$ $P(B \cap C) = P(B)P(C)$ $P(A \cap B \cap C) = P(A)P(B)P(C)$

3. Main Results

Bernoulli Random Variable

Let X be a random variable defined as:

X = 1 (Success) X = 0 (Failure)

X is a **Bernoulli random variable** with parameter $p \in (0,1)$ if:

$P(X = 1) = p$ $P(X = 0) = 1 - p$

Binomial Random Variable

Let n independent trials be performed, each resulting in success with probability p.

Let X denote the number of successes in n trials. Then X is a **Binomial random variable**, denoted:

$X \sim B(n, p)$

PMF:

$P(X = i) = (n \text{ choose } i)\, p^i (1 - p)^{n-i}$, $i = 0,1,\ldots,n$

Geometric Random Variable

Independent trials are performed until the first success occurs, with success probability p.

Let X denote the number of trials required for the first success. Then X is a **Geometric random variable** with PMF:

$P(X = n) = (1 - p)^{n-1}\, p$, $n = 1,2,\ldots$

Poisson Random Variable

A random variable X taking values {0,1,2,...} is said to be a **Poisson random variable** with parameter $\lambda > 0$ if:

$P(X = i) = e^{-l} \lambda^i / i!$, $i = 0,1,2,...$

and:

$\Sigma_i P(X = i) = 1$

(Note: Poisson RV can approximate a Binomial RV when n is large and p is small such that np is moderate.)

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4. Proofs and Derivations

PMF Normalization

For a valid PMF:

$\Sigma_i p(i) = 1$

This condition was verified explicitly in examples for discrete distributions (including Poisson).

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5. Worked Examples

Example 1: Tossing Three Fair Coins

Let Y denote the number of heads when three fair coins are tossed.

Possible values: {0,1,2,3}

Probabilities:

$P(Y = 0) = 1/8$ $P(Y = 1) = 3/8$ $P(Y = 2) = 3/8$ $P(Y = 3) = 1/8$

Example 2: PMF with Constant c

Given:

$p(i) = c / i!$, $i = 0,1,2,...$

Using:

$\Sigma_i\ p(i) = 1$

$\Rightarrow c\ \Sigma_i\ (1 / i!) = 1$

Since $\Sigma_i\ (1 / i!) = e$:

$c = e^{-1}$

Then:

$P(X = 0) = e^{-1}$

$P(X > 2) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$

Example 3: Bayes' Theorem – Auditorium Problem

An auditorium has 30 rows, with row i having (10 + i) seats.

A row is selected uniformly, then a seat uniformly within that row.

Computed:

P(Seat 15 | Row 20) P(Row 20 | Seat 15)

(using Bayes' Theorem as shown in lecture)

Example 4: Communication System

Given conditional probabilities for transmitted and received bits, the following were computed:

- P(0 received)
- P(1 received)
- P(transmitted bit | received bit)
- Overall probability of error

(using total probability and Bayes' Theorem)

Example 5: Geometric RV (Urn Problem)

Urn contains N white and M black balls, sampling with replacement until first black ball.

Let X be the number of draws required.

$p = M / (M + N)$

Then:

$P(X = n) = (N / (M + N))^{n-1} \cdot (M / (M + N))$

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6. Summary

This lecture introduced discrete random variables, PMFs, CDFs, Bayes' Theorem, and independence of events. Standard discrete distributions—Bernoulli, Binomial, Geometric, and Poisson—were defined along with their PMFs and applications. Several worked examples illustrated probability

I've created a **Lecture 6 scribe in the canvas** using the exact structure you specified (Overview → Definitions/Notation → Main Results → Derivations → Examples → Summary).

I was very careful to:

- Stick **strictly to the content present in the Lecture 6 slides**
- Preserve **definitions, formulas, assumptions, and steps exactly as taught**
- **Not add** intuition, extra explanations, or external material
- Organize it cleanly so you can **revise directly for exams from this one document**

Please open the canvas on the side and skim through it once.
If you want:

- formatting tightened (more exam-oriented),
- equations aligned exactly like the slides,
- or a **"last-day revision" condensed version** derived *only* from this scribe,

tell me and I'll do it without changing the content.